MTA Daily Ridership Analysis & Forecasting

Software Requirements Specification (SRS)

Initiative: DEPI - Data Analysis Track - Phase 1
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Executive Summary & Document Scope

This Software Requirements Specification (SRS) defines the business, functional, and non-functional requirements for the **MTA Daily Ridership Analysis & Forecasting** Capstone Project. Utilizing a comprehensive dataset of 1,706 daily records (spanning March 1, 2020, to October 31, 2024, across 7 transit modes), this project aims to bridge the gap between raw public transit data and executive decision-making. The final deliverable is an interactive Power BI dashboard that enables MTA executives and service planners to track recovery against pre-pandemic levels, isolate transit behaviors (e.g., Weekday vs. Weekend), and forecast total passenger ridership for the month following the last observation (November 2024) with a target Mean Absolute Percentage Error (MAPE) of 10% or less. Throughout this document, "total passenger ridership" means Subways, Buses, LIRR, Metro-North, and Staten Island Railway only; Access-A-Ride (scheduled trips) and Bridges & Tunnels (vehicle traffic) are reported separately.

1. Expanded Stakeholder Analysis

Stakeholder Role	Key Business Objectives / Needs	Impact / Influence	Engagement Strategy
MTA Executive Board	High-level strategic insights on overall recovery and future forecasts to drive budgeting and macro-policy decisions.	High	Provide a high-level KPI scorecard and executive summary dashboard page.

Stakeholder Role	Key Business Objectives / Needs	Impact / Influence	Engagement Strategy
Transit Service Planners	Granular trend analysis across all 7 modes to prioritize fleet capacity, manage routes, and address lagging services.	High	Deliver mode-specific drill-downs and pre-pandemic recovery percentage benchmarks.
Operations Managers	Day-to-day operational patterns (Weekday vs. Weekend) to optimize staff scheduling and maintenance.	Medium	Supply day-of-week analysis and seasonality visualizations.
Data & BI Team (Authors)	Clean, validated datasets with strict separation of passenger metrics from vehicular/scheduled metrics.	High	Implement strict data governance, Power Query ETL documentation, and GitHub version control.
DEPI Evaluation Committee	Proof of proper Star Schema modeling, accurate DAX calculations, and adherence to project constraints.	High	Ensure full RTM traceability, documented code, and project files submitted on time.
Commuters / Public Reps	Indirect beneficiaries of optimized service schedules and transparency in public transit recovery.	Low	Focus on data accuracy so downstream decisions positively impact public transit reliability.

2. User Stories & Acceptance Criteria

- **US-01 (Must):** As an MTA Executive, I want to track the overall daily and monthly passenger ridership trends since March 2020, so that I can evaluate the macro-level recovery progress.
    - *Acceptance Criteria 1:* Given the dashboard is open, a line or area chart must display total historical passenger ridership (see US-04) aggregated accurately over time.

    - *Acceptance Criteria 2:* When Year or Month slicers are adjusted, the visual must instantly filter to reflect the specified timeframe.
- **US-02 (Must):** As a Service Planner, I want to compare the "% of Comparable Pre-Pandemic Day" across all 7 modes, so that I can pinpoint lagging and recovering services.
    - *Acceptance Criteria 1:* A dedicated visualization must plot the recovery percentage for each mode individually (percentages are unit-free, so all 7 modes can be compared).
    - *Acceptance Criteria 2:* A static reference line at 100% (a value of 100 in the source percentage field) must be present to establish the pre-pandemic baseline.
- **US-03 (Must):** As an Executive, I want a data-driven forecast of total passenger ridership for November 2024, so that I can proactively adjust operational budgets.
    - *Acceptance Criteria 1:* The dashboard must output a calculated forecasted total for November 2024.
    - *Acceptance Criteria 2:* The forecast's Mean Absolute Percentage Error (MAPE) must be explicitly displayed via a backtest card, with a target of ≤ 10%. If the measured MAPE exceeds the target, the value is still displayed and the gap is documented. The MAPE is measured on daily values over a backtest period (August–October 2024), and the November 2024 total is the sum of the daily forecasts.
- **US-04 (Must):** As a Data Analyst, I want to ensure strict segregation of passenger ridership from Bridges & Tunnels vehicles and Access-A-Ride scheduled trips, so that aggregate KPIs are not mathematically skewed.
    - *Acceptance Criteria 1:* "Total Passenger" DAX measures must explicitly exclude Bridges & Tunnels and Access-A-Ride data rows.
    - *Acceptance Criteria 2:* Vehicular and scheduled trip data must be displayed using isolated, distinct calculation measures.
- **US-05 (Must):** As an Operations Manager, I want to analyze the variance between Weekday and Weekend average ridership, so that I can scale workforce deployment appropriately.
    - *Acceptance Criteria 1:* A visual component must calculate and compare the average ridership segmented by day-of-week classification (Weekday vs. Weekend).
    - *Acceptance Criteria 2:* This variance must update dynamically when filtered by Transit Mode.
- **US-06 (Should):** As an Executive, I want to monitor the Year-over-Year (YoY) ridership growth percentage, so that I can assess annual systemic performance.
    - *Acceptance Criteria 1:* A dynamic KPI card must render the YoY growth percentage.
    - *Acceptance Criteria 2:* The YoY calculation must appropriately handle leap years or date disparities using standard DAX Time Intelligence.

- **US-07 (Must):** As a Service Planner, I want data anomalies (e.g., the zero-value days in the Staten Island Railway fields) handled transparently, so that average calculations remain statistically valid.
    - *Acceptance Criteria 1:* Zero-values in the Staten Island Railway ridership field (32 days, 21 June 2020 - 15 August 2021) and in its pre-pandemic percentage field (54 days, 30 May 2020 - 30 October 2022) must be reviewed and, where they represent missing observations, intercepted in Power Query (e.g., converted to nulls) to prevent skewing average metrics.
    - *Acceptance Criteria 2:* The anomaly handling procedure must be logged in the project documentation.
- **US-08 (Must):** As a User, I want global dynamic interactivity and cross-filtering, so that I can explore intersections between Date, Mode, and Day Type across all pages.
    - *Acceptance Criteria 1:* Global slicers for 'Date (Year/Month)' and 'Transit Mode' must be accessible on a unified filter pane or prominently on each page.
    - *Acceptance Criteria 2:* Selections made in one visual must accurately cross-filter all other visuals within the same page context.
- **US-09 (Should):** As an MTA Executive, I want a Performance & KPI Scorecard prominently displayed, so that I can grasp high-level systemic health at a glance.
    - *Acceptance Criteria 1:* The top of the dashboard must feature a consolidated row of KPI cards (Total Passenger Ridership, Overall Recovery ⁄ weighted across the five passenger modes, November Forecast).
- **US-10 (Must):** As a DEPI Evaluator, I want full traceability, quality documentation, and version control, so that I can audit the ETL logic and dataset lineage.
    - *Acceptance Criteria 1:* A comprehensive README.md file must be hosted on the GitHub repository detailing the Power Query steps and DAX dictionary.
- **US-11 (Must):** As a Service Planner, I want to identify the seasonal peaks and troughs in passenger ridership, including the top 3 peak days, so that I can align service capacity with recurring demand.
    - *Acceptance Criteria 1:* A chart must show average passenger ridership by calendar month, making the highest and lowest months identifiable.
    - *Acceptance Criteria 2:* A table or visual must list the top 3 peak days by total passenger ridership for the selected period and update when filtered by Transit Mode.

3. Functional Requirements (FR)

- **FR-01 (Data Ingestion & ETL):** The system shall ingest the 1,706-row dataset (7 transit modes) and apply Power Query ETL processes, including null substitution for validated Staten Island anomalies, to produce a clean analytical table.
- **FR-02 (Metric Segregation):** The system shall compute "Total Passengers" (Subways, Buses, LIRR, Metro-North, and Staten Island Railway) via DAX, strictly isolating

standard transit modes from non-passenger categories (Access-A-Ride, Bridges & Tunnels).

- **FR-03 (Baseline Comparison):** The system shall calculate and visualize the "% of Comparable Pre-Pandemic Day" using a static 100% reference line benchmark.
- **FR-04 (Forecasting & MAPE):** The system shall execute a forecasting algorithm to predict November 2024 total passenger ridership and calculate the associated Mean Absolute Percentage Error (MAPE) against a backtest subset (daily values, August–October 2024); the November 2024 total is the sum of the daily forecasts.
- **FR-05 (Time Intelligence DAX):** The system shall utilize DAX Time Intelligence functions to compute Year-over-Year (YoY) growth and segment aggregations by Weekday vs. Weekend.
- **FR-06 (Global Filtering/Slicers):** The system shall provide synchronized global slicers allowing multifaceted filtering across Date hierarchies and Transit Modes.
- **FR-07 (Executive KPI Rendering):** The system shall render a dedicated Executive Scorecard featuring high-level aggregates updated dynamically by report context.
- **FR-08 (Data Lineage & Quality Logging):** The system shall support transparent documentation of data lineage, ensuring all cleaning steps are verifiable.
- **FR-09 (Star Schema Data Model):** The system shall provide a validated Star Schema with one fact table and Date and Transit Mode dimensions, with relationships verified against the source data.
- **FR-10 (Seasonality & Peak Analysis):** The system shall display average passenger ridership by calendar month and list the top 3 peak days for the selected period and Transit Mode.
- **FR-11 (Dashboard Structure & Navigation):** The system shall provide at least four analytical pages (executive overview, recovery analysis, patterns and seasonality, and forecast) with consistent navigation between them.

4. Non-Functional Requirements (NFR)

- **NFR-01 (Performance & Responsiveness):** The underlying Star Schema and DAX formulas must be optimized such that cross-filtering and visual updates execute in under 3 seconds, as measured with Power BI Performance Analyzer.
- **NFR-02 (Data Accuracy & Aggregation Integrity):** Dashboard aggregated totals must maintain 100% consistency with the source CSV file after the documented cleaning rules are applied (null handling for Staten Island anomalies and exclusion of non-passenger modes from passenger totals).
- **NFR-03 (Usability, Accessibility & UI/UX):** The UI shall use one consistent professional theme (navy primary color #1F3864) with high-contrast text and a colorblind-safe palette for categorical series, applying the same color to each transit mode on every page.
- **NFR-04 (Traceability & GitHub Documentation):** The project repository must include a

comprehensive README.md, capturing step-by-step Power Query ETL logs and the DAX data dictionary.

- **NFR-05 (Display Compatibility):** The dashboard canvas shall use a 16:9 layout (1280×720, the Power BI default) for consistent desktop viewing; the same ratio also suits potential web embedding.

5. Requirements Traceability Matrix (RTM)

User Story	Mapped FR	Mapped NFR	Feature Component	Target Verification Test Case
US-01 (Macro Trend)	FR-01, FR-02, FR-06, FR-09	NFR-01, NFR-02	Line Chart & Date Slicers	Verify visual updates within 3s; passenger totals match raw dataset grouping.
US-02 (Parity)	FR-03	NFR-02, NFR-03	Bar/Line Chart with 100% Line	Check if Y-axis scales correctly and 100% baseline is visible.
US-03 (Forecast)	FR-02, FR-04	NFR-02	Forecast Visual & MAPE Card	Confirm the Nov '24 value is generated and the MAPE card is displayed (target ≤ 10%).
US-04 (Data Separation)	FR-02	NFR-02	DAX Measure Architecture	Sum of "Total Passengers" excludes Traffic/Access-A-Ride.
US-05 (Weekday vs. Weekend)	FR-05	NFR-01	Matrix or Column Chart	Weekday/Weekend ratio matches manual Excel calculation.
US-06 (YoY Growth)	FR-05	NFR-02	KPI Card	Select year 2023; verify YoY % aligns with 2022 raw data comparison.

User Story	Mapped FR	Mapped NFR	Feature Component	Target Verification Test Case
US-07 (Anomalies)	FR-01, FR-08	NFR-02, NFR-04	Power Query Editor	Filter Staten Island; ensure zeroes do not drag down valid day averages.
US-08 (Interactivity)	FR-06, FR-11	NFR-01, NFR-05	Unified Slicer Pane	Select "Buses"; verify all charts refresh in < 3s without layout breaks.
US-09 (Scorecard)	FR-02, FR-07, FR-11	NFR-03, NFR-05	Top KPI Banner	Cards are readable, color-safe, and fit within 16:9 layout.
US-10 (Traceability)	FR-08	NFR-04	GitHub Repository	README.md is populated with ETL steps and DAX measures.
US-11 (Seasonality & Peaks)	FR-10	NFR-01, NFR-02	Monthly Column Chart & Top-3 Peak Days Table	Highest/lowest months and top 3 peak days match a manual Excel calculation.